Amber Fredenburg, PhD

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I am an ecologist interested in how communities are influenced by environmental change. My areas of expertise include plant–pollinator interactions, conservation biology, biodiversity monitoring, eDNA techniques, field experiments, observational studies, and native bee ecology and natural history.

Education

PhD The Ohio State University, Columbus, Ohio — 2026
Evolution, Ecology, and Organismal Biology
Dissertation: *Digging deeper: Unearthing the nesting requirements of ground-nesting bees*
Graduate Minor, Public Policy

MS Lakehead University, Thunder Bay, Ontario — 2020
Forestry
Thesis: *Diversity, phenology, and host associations of wild bees (Hymenoptera: Anthophila) in Thunder Bay, Ontario*

BS The Ohio State University, Columbus, Ohio — 2018
Zoology

Professional Experience

Senior Research Associate — The Ohio State University — 2026-Present
Department of Evolution, Ecology, and Organismal Biology
The Limits of eDNA for Monitoring Endangered Pollinators
Funded by the United States Department of Defense

- Piloted the collection of native ground-nesting bee eDNA samples from nests and foraging pollinator eDNA samples from leaves and flowers.
- Designed and implemented DNA extraction methods for eDNA samples to ensure sample purity and target taxa amplification
- Prepared and performed PCR and gel electrophoresis techniques on eDNA samples, ensuring the success of multiple primers in amplifying different insect taxa.

Graduate Research Assistant — The Ohio State University — 2022–2026
Department of Evolution, Ecology, and Organismal Biology
The Ohio Bee Survey and *Specialist Bee Project*
Funded by the Ohio Department of Natural Resources

- Managed teams of researchers and community scientists to locate and collect data on wild bee populations across Ohio using standardized field sampling and spatial analysis.
- Designed and completed four research studies, investigating large-scale soil influence on bee community structure, local habitat associations of diet specialist bee populations, applied conservation strategies for enhancing nesting habitat, and the statewide distribution of Ohio's bee species.
- Collaborated with state division of wildlife agencies to translate survey data into listed species of conservation concern and monitoring recommendations.
- Assembled and organized large datasets, conducted statistical and spatial analyses in R and GIS, and wrote manuscripts for publication.

Graduate Teaching Assistant — Ohio State University — 2021–2024
Department of Evolution, Ecology, and Organismal Biology; Center for Life Sciences Education

- Led lab sections for undergraduate students in Biology, Ecology, and Organismal Biology courses.
- Guided students in the interpretation of lecture material, dissection, experimental design for independent projects, and data analysis.
- Graded assignments, lab reports, and exams while providing feedback.

- Assisted with course preparation and coordination with faculty.
- Supported students with office hours and one-on-one mentoring.

Graduate Teaching Assistant — Lakehead University 2019–2020

Department of Natural Resources Management

- Assisted instruction and led lab sections for undergraduate students in Forest Entomology and Population Ecology.
- Supported lab exercises involving insect identification, specimen handling, and data analysis.
- Graded assignments and exams while providing feedback.
- Assisted with course preparation and coordination with faculty.
- Supported students with office hours and one-on-one mentoring.

Undergraduate Research Assistant — The Ohio State University 2017–2018

Department of Evolution, Ecology, and Organismal Biology

Pumpkin Pollinator Project and *Ohio Bumble Bee Survey*

Funded by the Ohio Department of Transportation

- Conducted statewide field surveys to detect bumble bee populations across Ohio, documenting species distributions, abundance, and habitat associations.
- Implemented standardized field methods for population monitoring, including visual identification and ecological data collection across multiple sites.
- Recorded and managed field-based datasets to support analysis of population trends and environmental associations.

Publications

Fredenburg, A. and K. Goodell. (2026). Soil sanctuaries: Experimental manipulations enhance ground-nesting bee habitat. *Ecosphere,* 17(10): e70781. https://doi.org/10.1002/ecs2.70781

Fredenburg, A. (2026). *Digging deeper: Unearthing the nesting requirements of ground-nesting bees* [Doctoral dissertation, Ohio State University]. OhioLINK Electronic Theses and Dissertations Center. http://rave.ohiolink.edu/etdc/view?acc_num=osu1776083021090548

Pen, I. A. M., Dueñas, L. F., Armondo Sánchez, J., Vergera-Flórez, D. C., Gómez, C. E., Tovar-Bolaños, V., Palacios, J. A., Moreno, J. A., Stoilova, M. I., Lo, C., Kephart, C., Fredenburg, A., Muñoz, R. G., Simões, N., Lizaloa-Gullermo, J. & Daly, M. (2026). New reports of Bunodeopsis Andres, 1881 in the Greater Antilles, Southern, Southeastern, and Western Caribbean. *Marine Ecology.* 47(2): e70081.https://doi.org/10.1111/maec.70081

Fredenburg, A. N., J. Gibbs, and D. C. Henne. (2021). A survey of wild bees (Hymenoptera: Anthophila) and their floral associations in Thunder Bay, Ontario. *Journal of the Kansas Entomological Society.* 93(3): 211-229. https://doi.org/10.2317/0022-8567-93.3.211

Fredenburg. A., Spring, M., and K. Goodell. Glacial history structures the soil characteristics associated with ground-nesting bee richness, species composition, and traits. (in review).

Fredenburg. A., Spring, M., and K. Goodell. The role of nesting resources on specialist bee populations and the bee community (in prep).

Goodell, K., Fredenburg, A., Spring, M., and Semler, N. The bees of Ohio: a statewide checklist of species (in prep).

Presentations and Posters

2026 **Fredenburg, A.**, Spring, M, & Goodell, K—Glacial history structures the soil characteristics associated with ground-nesting bee richness, species composition, and traits. *Entomological Society of America*, Columbus, Ohio.

Sakulich, L., **Fredenburg, A.**, & Goodell, K—Building networks between plants, pollinators, and students in the classroom using an interactive lab activity. *Entomological Society of America*, Columbus, Ohio

Fredenburg, A.—Digging deeper: Unearthing the resource requirements of ground-nesting bees, *Evolution, Ecology, and Organismal Biology Departmental Seminar,* Ohio State University, Columbus, Ohio.

2025 **Fredenburg, A.**, Goodell, K., Spring, M., & Semler, N.—Capturing Ohio's bee diversity by surveying the host plants of diet specialists, *invited speaker, *Oak Openings Science Summit*, Toledo, Ohio

Fredenburg, A. & Goodell, K.—Bee responses to resources depend on ecological scale, *Ecological Society of America*, Baltimore, Maryland

2024 **Fredenburg, A.**, & Goodell, K.—Exploring the link between soil characteristics and the bee across Ohio, *Natural Resource Research and Management Symposium*, Cleveland, Ohio

Goodell, K., Spring, M., **Fredenburg, A.**, Semler, N., & Fetters Fogarty, A.—The Ohio Bee Survey: species checklist and planning future monitoring, *Natural Resource Research and Management Symposium*, Cleveland, Ohio

Fredenburg, A. & Goodell, K.—Enhancing the nesting habitat for ground-nesting bees, *invited symposium speaker, *Entomological Society of America*, Phoenix, Arizona

Goodell, K., **Fredenburg, A.**, Semler, N., & Spring, M.—Optimizing wild bee surveys for conservation management: a comparison of survey methods, *Entomological Society of America,* Phoenix, Arizona

2023 **Fredenburg, A.**, Spring, M., & Goodell, K.—Documenting specialist bee diversity and factors that influence their distributions in Ohio, *Ohio Natural History Conference*, Cincinnati, Ohio

Semler, N., Spring, M., **Fredenburg, A.**, & Goodell, K.—Native and non-native thistle (*Cirsium* spp.) support many generalists, but fewer specialist bees, *Ohio Invasive Plant Research Conference,* Columbus, Ohio

Fredenburg, A., Spring, M., & Goodell, K.—The role of nesting resources on bee abundance at host plant patches, *Entomological Society of America*, National Harbor, Maryland

Goodell, K., Spring, M., Helton, C., & **Fredenburg, A**., Semler, N., & Spring, M.—Lessons from the Ohio Bee Survey: Analysis of pan traps and suggestions for improving specimen richness while limiting specimen numbers, *Entomological Society of America,* National Harbor, Maryland

2022 Goodell, K., Spring, M., Helton, C., & **Fredenburg, A.**—The role of midwestern USA forests in Supporting bee populations, *Entomological Society of America*, Vancouver, British Columbia

2020 **Fredenburg, A.**, & Henne, D.—Native bee composition and diversity at three sites in Thunder Bay, Ontario, *Lakehead University Graduate Studies Thesis Presentations,* Thunder Bay, Ontario

Fredenburg, A., & Henne, D.—Biodiversity of native bees at three sites in Thunder Bay, Ontario, *Graduate Studies Research Conference Poster Competition* (Winner), Thunder Bay, Ontario

Students Mentored

Ohio State University

Gianna Campbell, *The Limits of eDNA for Monitoring Endangered Pollinators* (2026)

Hamzah Albairmani, *The Limits of eDNA for Monitoring Endangered Pollinators* (2026)

Rex Harvey-Thurston, *Ground-Nesting Plot Project* and *Specialist Bee Project* (2023-2026)

Abby Subler, *Specialist Bee Project* (2024-2025)

Noma Semler, *Specialist Bee Nesting Project* (2022-2023)

Vanessa Chilcoat, *Diversity and Distribution of Asilidae of Ohio* (2022)

Lakehead University

Alena Albertson-Kok, *Diversity of Saproxylic Beetles in Thunder Bay, ON* (2019-2020)

Community Engagement

Convening Conservationists Talk, Great Parks of Hamilton County, Cincinnati, Ohio (4/21/2026) – disseminated Ohio Bee Survey results and discussed guidelines for developing a bee monitoring program and how to construct effective pollinator habitat.

Science Olympiad Entomology B, (October 2024-April 2025)—wrote the Entomology exam with a team of co-authors from Ohio State University for middle school and high school students participating in Science Olympiad 2025.

Franklin County Pollinator Pathways, Franklin Soil and Water Conservation, Columbus, Ohio (6/6/2024)—invited as a guest bee expert for a gathering of representatives, conservation groups, advocacy groups, and other organizations to discuss how to create more and less fragmented native pollinator habitat in Franklin County.

FLOW Webinar: Creating habitat for our native ground-nesting bees, Friends of the Lower Olentangy Watershed, Columbus, Ohio (5/20/2024)—lead a webinar for members and anyone interested in creating ground-nesting bee habitat.

FLOW Science Board Meeting, Friends of the Lower Olentangy Watershed, Columbus, Ohio (1/8/2024)—discuss research results with the science committee of FLOW and discuss ways to improve pollinator gardens for nesting bees.

Native Bees and Where They Nest, Beecoming Home, The Bee Collective, Columbus, Ohio (9/9/2023)—engaged the public with information about native bee nesting at a collaborative art and science event.

Science Day at Parkmoor Elementary School, Scientific Thinkers, Ohio State University (12/8/2022), —lead volunteer instructor for a strawberry DNA removal experiment in 3rd grade classrooms.

Bees, I Love Them, and You Should Too—presentation to engage K-12 students in the importance of bee conservation, Driscoll Elementary, Centerville City Schools, Centerville, Ohio, (4/15/2020) and Fredrick Douglas Elementary, Cincinnati City Schools, Cincinnati, Ohio (5/4/2023)

Professional Service

Biodiversity and Conservation – Reviewer (2026, 1 manuscript)

Scientific Reports – Reviewer (2026, 1 manuscript)

Journal of Insect Behavior – Reviewer (2025, 1 manuscript)

Journal of Insect Conservation – Reviewer (2025, 1 manuscript)

Journal of Applied Ecology – Reviewer (2024, 2 manuscripts)

Journal of Restoration Ecology – Reviewer (2022-2023, 2 manuscripts)

Ecology and Evolution – Reviewer (2022, 1 manuscript)

Grants Received

2023 Garden Club of America Pollinator Fellowship. $4,000.

Beyond Blooms: Enhancing nesting resources for ground-nesting bee conservation

2021 ENGIE-Axium Research Incentive Recruitment Scholarship Award. $5,000.

Skills

Analytical & Computational

- R, ArcGIS, statistical modeling, multivariate analyses, data visualization

Laboratory

- DNA extractions, PCR, gel electrophoresis

Field

- Experimental and observational study design, insect and environmental surveys, field coordination, data management

Specimen Collections

- Insect specimen preparation, pinning, labeling, identifying, databasing, and collection organization
- Bee species identification